Aryan Tembhekar

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PROFESSIONAL SUMMARY

Data-driven Supply Chain Analyst and Analytics Engineer with a Mechanical Engineering foundation from SVNIT Surat. Proven expertise in architecting enterprise-scale manufacturing analytics platforms and automated reporting systems processing 3M+ records. Highly skilled in bridging operational logistics with full-stack data solutions using Python, SQL, DuckDB, Node.js, and React to drive instant, high-level management decision-making.

TECHNICAL SKILLS

Data & Analytics: Data Pipeline Development, Manufacturing Analytics, Statistical Analysis, Automated Reporting, Dashboard Development, Historical vs. Live Production Modeling
Programming Languages: Python, SQL, JavaScript, HTML, CSS
Databases & Platforms: DuckDB, Relational SQL Databases, Node.js, React
Tools & Enterprise Systems: Git (Gitea), Advanced Microsoft Excel, VS Code, Linux Basics, SAP S/4HANA Analytics

PROFESSIONAL EXPERIENCE

Shree Ram Krishna Exports Pvt. Ltd. — **Surat, India**
Supply Chain Analyst – Data & Manufacturing Analytics — June 2025 – Present

- Architected a manufacturing analytics platform processing 3M+ production records across 92 operational parameters, encompassing rough-to-polish diamond data from 2021 to present.
- Engineered an automated reporting infrastructure that completely eliminated manual data compilation, reducing reporting time from 30–45 minutes to instant generation for 200–300 daily operational reports used by senior management.
- Developed high-performance centralized data repositories utilizing DuckDB for rapid querying and analysis of massive manufacturing datasets.
- Built end-to-end web analytics dashboards using React and JavaScript, powered by custom Node.js backend services and internal APIs, to visualize critical production trends and logistics metrics.
- Implemented statistical analysis workflows in Python to compare historical and live production data, directly optimizing polishing decision-making processes.
- Designed a comprehensive packet tracking interface and automated inventory tools to monitor diamond movement and operational throughput across manufacturing stages involving 9,000+ employees.

RESEARCH & LEADERSHIP EXPERIENCE

Indian Institute of Technology (IIT) Mandi — **Mandi, India**
Research Intern — June 2024 – July 2024

- Designed a teleoperated robotic arm mounted under UAV platforms for specialized aerial operations.
- Executed system integration and component selection for autonomous ground vehicles (AGVs) aiming for railway anomalies.

SAE Phoenix Aero — **Project Manager**

- Led a multidisciplinary engineering team in the development of UAV systems for national competitions, overseeing both hardware design and software logic.
- Organized and led technical workshops focused on drone systems and autonomous navigation technologies.

ASHINE Incubator — **Innovation Lead**

- Secured INR 1,00,000 in innovation funding to develop an autonomous drone delivery platform.

ENGINEERING PROJECTS

- **Autonomous Drone Development Platform:** Designed an aerodynamically optimized quadrotor UAV. Integrated a Raspberry Pi computing module with ArUco marker detection for GPS-independent precision landing. Implemented an IoT payload monitoring system (temperature, humidity, pressure). **Achievement:** Top 10 national ranking in SAE Autonomous Drone Development Challenge.
- **Hexapod Autonomous Robot:** Engineered an 18-DOF robotic hexapod platform capable of autonomous navigation and object detection. Developed custom gait algorithms for improved mobility across uneven terrain. **Achievement:** 2nd Overall Rank at Robofest robotics competition.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT) — **Surat, India**
Bachelor of Technology in Mechanical Engineering — May 2025 CGPA: 7.71/10

AWARDS & ACHIEVEMENTS

- Product Patent: Aerodynamically Stable Quadrotor Platform
- 3rd Prize: Innovative Quadrotor Design (SAE India)
- 2nd Rank: Robofest Hexapod Robotics Competition